Negative Sampling in Recommendation: What makes a Good Negative?

MINDS Lab.

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Contents

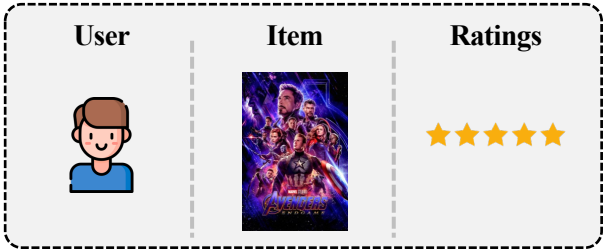
- Negative Sampling in Recommendation
- DNS: Mine Hard Negatives
- SRNS: Hard but Reliable Negatives
- DNS+: Why Does Hard Negative Sampling Work?

Recommendation Learns from User Feedback

Negative Sampling in Recommendation

Explicit Feedback

- Users directly express their preferences
  - Rating, Review, Like / Dislike, ...

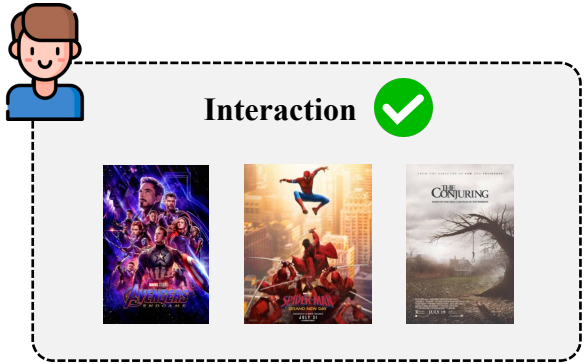


Implicit Feedback

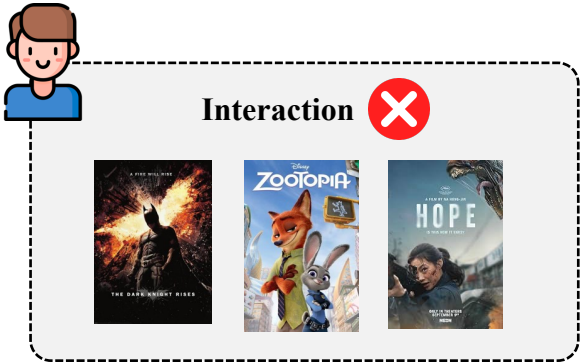
- Preferences are inferred from user behavior
  - Click, Purchase, Watch, Play, ...

Negative Samples under Implicit Feedback

❖ Where Do Negative Samples Come From?



Observed ➞ ***Positive***



Unobserved ➞

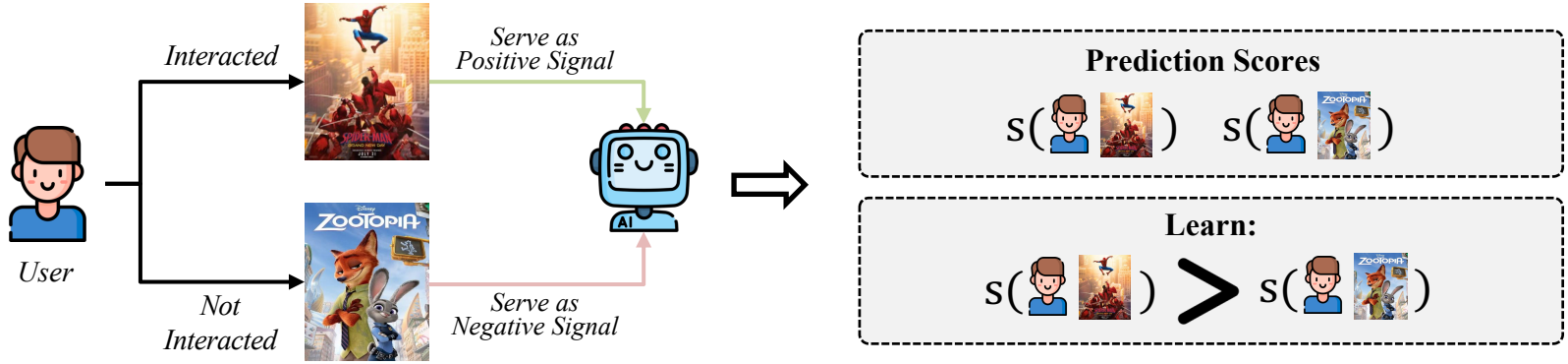
Negative Candidate Pool

❑ Negative sampling selects a subset of training negatives from the massive unobserved item pool

How Are Positive and Negative Samples Used in Training?

Negative Sampling in Recommendation

❖ Learning Relative Preference from Positive and Negative Samples

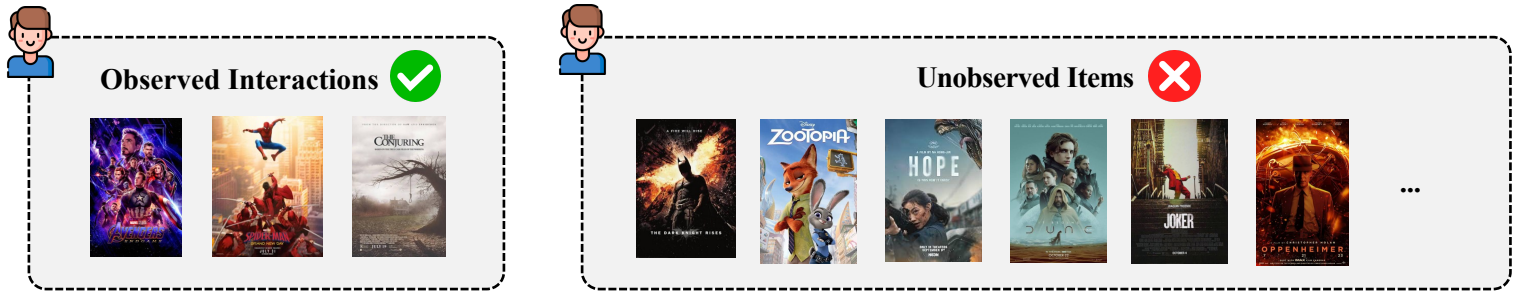


- Positive–Negative Comparison
  - Preference scores for positive and sampled negative items
- Relative Preference Optimization
  - Higher score for positives than sampled negatives

Why Do We Need Sampling?

Negative Sampling in Recommendation

❖ Too Many Negative Candidates for Training



❑ Large-scale Recommendation

- Millions of users & items
- Massive item corpus

❑ Sparse User Interactions

- Few observed interactions per user
- Most user–item pairs remain unobserved

❑ Computational Burden

- Full-corpus training is impractical
- High cost per training update

Uniform Negative Sampling

❖ The Simplest Sampling Strategy

- ❑ Model-Agnostic Selection
  - ▸ No consideration of current model predictions
- ❑ Simple and Efficient
  - ▸ No additional scoring or side information

From Hard Negatives to Better Negatives

Negative Sampling in Recommendation

- **How the definition of a “good negative” evolves**
  - DNS — Find Hard Negatives
    - **Good Negative = Hardness**
  - SRNS — Make Hard Negatives Reliable
    - **Good Negative = Hardness + Reliable**
  - DNS+ — Control the Right Hardness
    - **Good Negative = Appropriately Hardness**

Optimizing Top-N Collaborative Filtering via Dynamic Negative Item Sampling

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Do All Negatives Look the Same to the Current Model?

DNS: Mine Hard Negatives

❖ Same Sampling Rule, Different Model Responses

Same sampling probability

$$P(\text{마일드씽}) = P(\text{ZOOTOPIA}) = P(\text{THE DARK KNIGHT RISES})$$

Different model predictions

$$s(\text{마일드씽}) \neq s(\text{ZOOTOPIA}) \neq s(\text{THE DARK KNIGHT RISES})$$

Negative Samples

Positive Sample

0.05	0.35	0.78	0.82
Well seperated		*Poorly seperated*	*Positive*

Model Score

Illustrative model scores; higher scores indicate stronger predicted preference

- ❑ Different Levels of Separation
  - ▸ Same sampling probability, but different score gaps from the positive

➔ **Does this difference matter during training?**

Why Does the Negative Score Matter?

DNS: Mine Hard Negatives

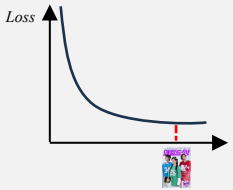
❖ Different Score Gaps, Different Optimization Effects

$$\mathcal{L}_{BPR} = -\log \sigma\left(s(u,i^+) - s(u,j^-)\right), \qquad s(u,i^+)\colon 0.82$$ 

Well-separated negative

$s(u,j_1) = 0.05$

$\Delta_1 = 0.82 - 0.05 = 0.77$

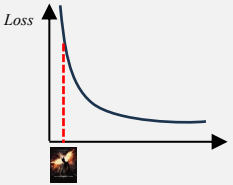


Smaller gradient magnitude

Poorly-sepearated negative

$s(u,j_2) = 0.78$

$\Delta_2 = 0.82 - 0.78 = 0.04$



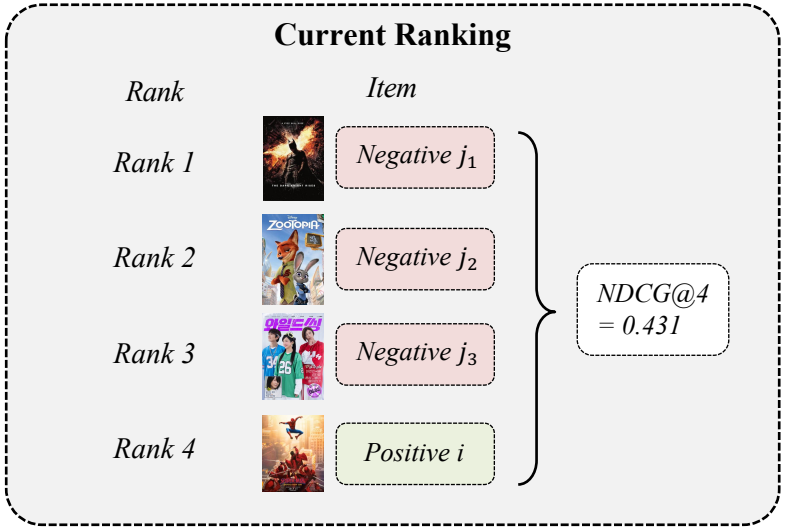
Larger gradient magnitude

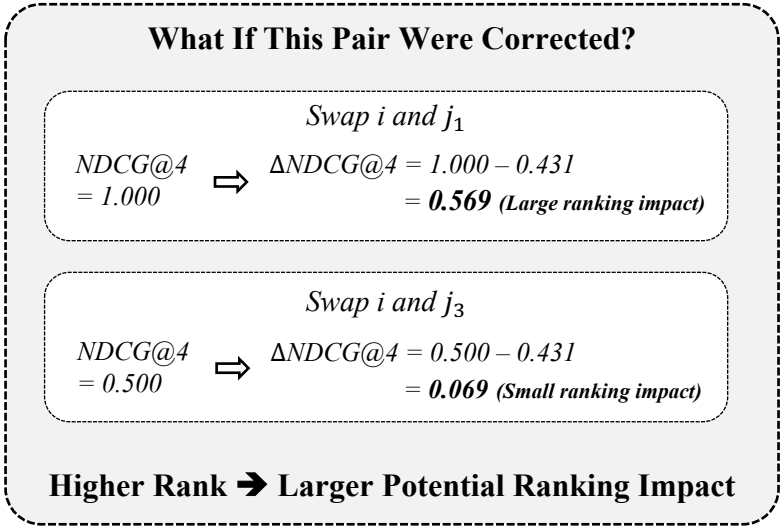
- Already well-separated negatives provide less additional optimization pressure
- The usefulness of a negative depends on the current model state

Rank Matters

DNS: Mine Hard Negatives

- **Top-N recommendation is evaluated by rank-biased metrics**





- Errors near the top matter more for rank-biased metrics

Weight What Matters

DNS: Mine Hard Negatives

❖ From ranking impact to learning priority

BPR

$$\frac{\partial L_{ij}}{\partial w} = \frac{\partial L_{ij}}{\partial (s_{ui} - s_{uj})} \left(\frac{\partial s_{ui}}{\partial w} - \frac{\partial s_{uj}}{\partial w} \right)$$

$$= \lambda_{ij} \left(\frac{\partial s_{ui}}{\partial w} - \frac{\partial s_{uj}}{\partial w} \right)$$

Pairwise Gradient Signal: λ_{ij}

"How difficult is this pair for current model?"

⇨

Rank-aware Weighting

Rank-aware Gradient Scale: $\lambda_{ij} \cdot \Delta NDCG_{ij}$

"How important is this error?"

$\Delta NDCG@4 =$ ***0.569*** ➔ ***Higher Priority***

$\Delta NDCG@4 =$ ***0.069*** ➔ ***Lower Priority***

- But full ranking is expensive
  - Computing $\Delta NDCG$ requires the current ranking over all unseen items

➔ **Can we get the same effect through sampling instead of explicit weighting?**

Sample What Matters

DNS: Mine Hard Negatives

From rank-aware weighting to rank-aware sampling

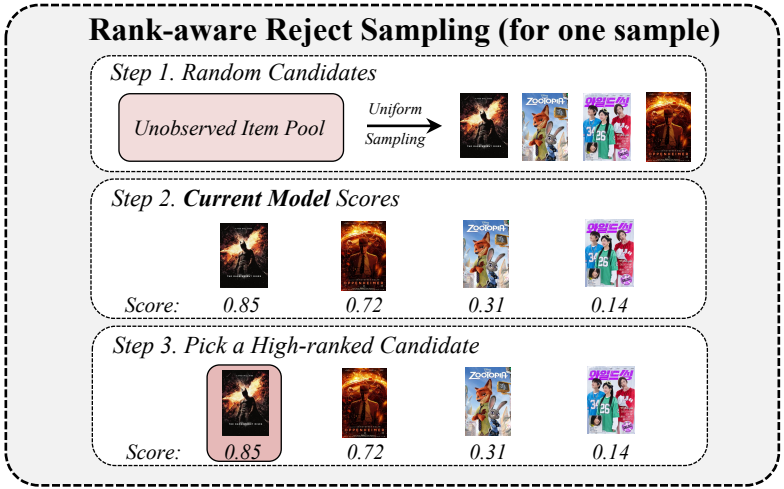
Sampling Instead of Weighting

$$p_j \propto \frac{f(\lambda_{ij}, \zeta_u)}{\lambda_{ij}}, f(\lambda_{ij}, \zeta_u) = \lambda_{ij} \cdot \Delta NDCG_{ij}$$

$$\rightarrow p_j \propto \Delta NDCG_{ij}$$

Larger ranking impact → Higher sampling probability

But ΔNDCG itself still requires full ranking...

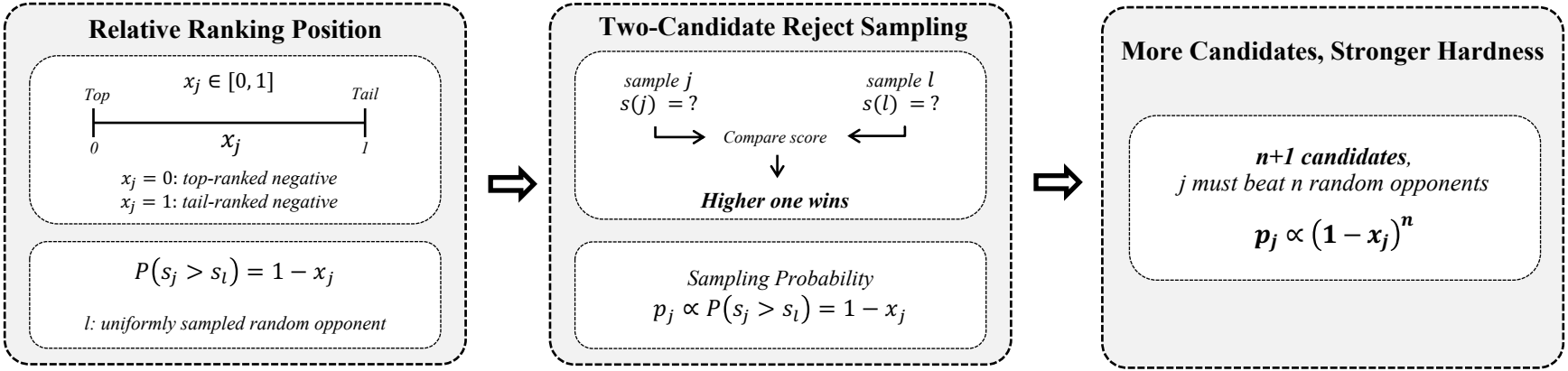


- Sample a few → Score them → Prefer higher-ranked negatives
  - Dynamic: the hard negative changes as the model changes

Sample What Matters (Cont.)

DNS: Mine Hard Negatives

❖ Why does reject sampling become rank-aware?



- ❑ Higher-ranked negatives win random comparisons more often ➔ Higher Sampling Probability
- ❑ Random competition implicitly approximates a rank-biased sampling distribution

Does DNS Improve Learning?

DNS: Mine Hard Negatives

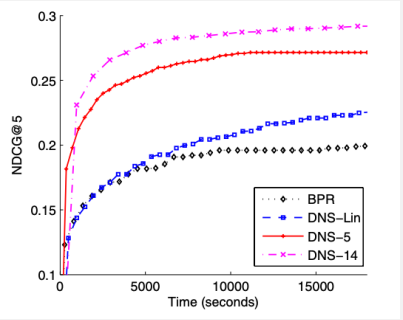
- **DNS improves both ranking quality and training efficiency**

Ranking Performance

Netflix					
	P@5	P@10	NDCG@5	NDCG@10	MAP
BPR	0.3826	0.3272	0.2052	0.2017	0.1403
DNS	0.4708	0.4012	0.2906	0.2887	0.2036
Impv.	23.1%*	22.6%*	41.6%*	43.1%*	45.1%*
Yahoo! Music					
	P@5	P@10	NDCG@5	NDCG@10	MAP
BPR	0.1588	0.1359	0.1683	0.1481	0.0615
DNS	0.4243	0.3671	0.4458	0.3981	0.1644
Impv.	167.2%*	170.1%*	164.9%*	168.8%*	167.3%*
Last.fm					
	P@5	P@10	NDCG@5	NDCG@10	MAP
BPR	0.1231	0.1168	0.1270	0.1207	0.0221
DNS	0.1323	0.1202	0.1355	0.1250	0.0223
Impv.	7.5%*	2.9%	6.7%*	3.6%	0.9%

Harder negatives substantially improve rank-biased metrics

Training Efficiency



More informative updates require fewer training rounds

- Current model score provides a useful signal for negative selection
- Sampling informative negatives improves both ranking quality and convergence

Simplify and Robustify Negative Sampling for Implicit Collaborative Filtering

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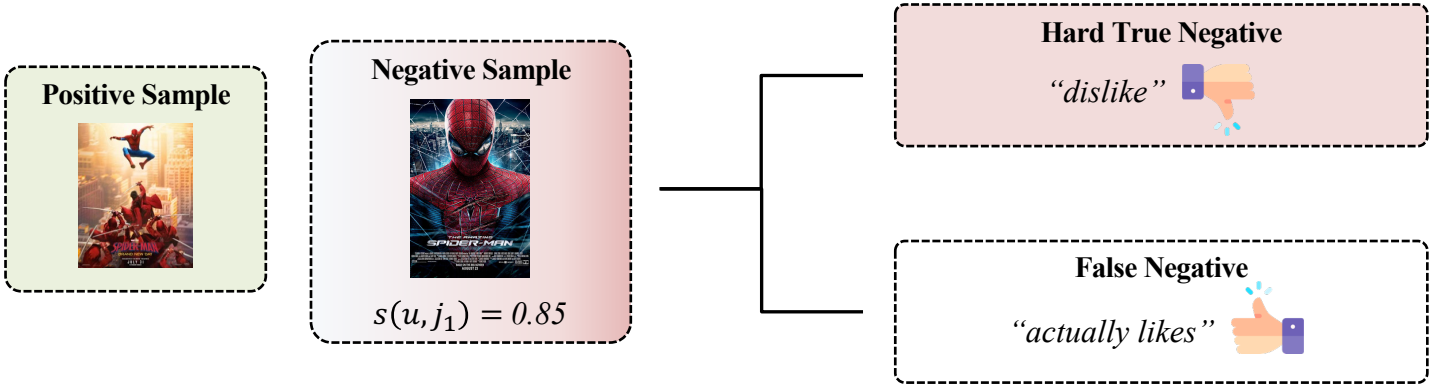
Is Hard Always Good?

SRNS: Hard but Reliable Negatives

- **Hard negative sampling may amplify false negatives**



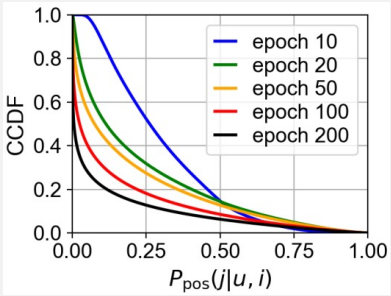
- An item may be unobserved because of exposure, position, or other factors — not necessarily dislike
  - High Score ≠ True Negative
- Can we distinguish hard true negatives from false negatives?

Understanding False Negatives

SRNS: Hard but Reliable Negatives

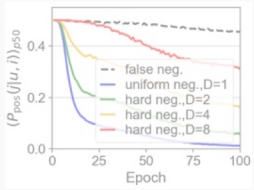
❖ Finding 1: Hard negatives are rare

Only a Few Negatives Are Hard



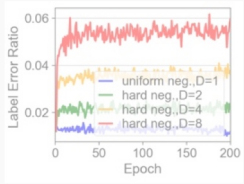
$$F(x) = P(P_{pos} \geq x)$$

False Negatives Also Look Hard



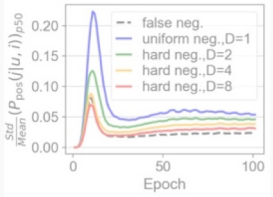
D: number of random candidates used to select one negative

Harder Sampling Selects More False Negatives



$$\text{Label Error Ratio} = \frac{\#\text{ selected false negatives}}{\#\text{ selected negatives}}$$

False Negatives Are More Stable



$$\text{Normalized Variance} = \frac{\text{Std}(P_{pos})}{\text{Mean}(P_{pos})}$$

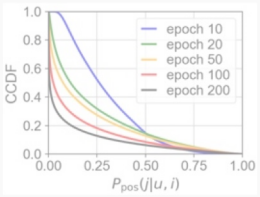
- Only a small fraction of unobserved items have high positive-label probability P_{pos}

Understanding False Negatives

SRNS: Hard but Reliable Negatives

❖ Finding 2: Score captures hardness, but not reliability

Only a Few Negatives Are Hard



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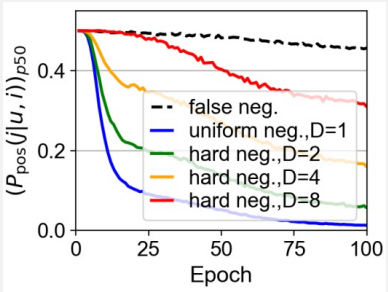

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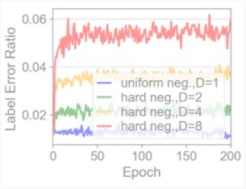
Harder Sampling Selects More False Negatives



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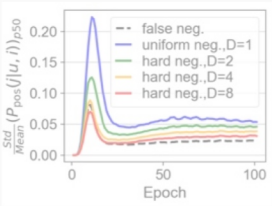
False Negatives Are More Stable



$\text{Normalized Variance} = \frac{Std(P_{pos})}{Mean(P_{pos})}$

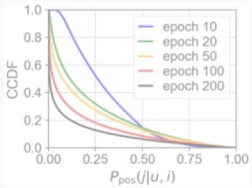
- Harder negatives have larger P_{pos}, but false negatives also maintain high scores
- Prediction score alone cannot distinguish hard true negatives from false negatives

Understanding False Negatives

SRNS: Hard but Reliable Negatives

- **Harder is more informative, but also noisier**

Only a Few Negatives Are Hard



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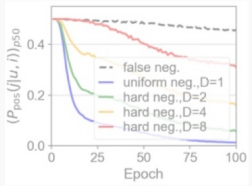

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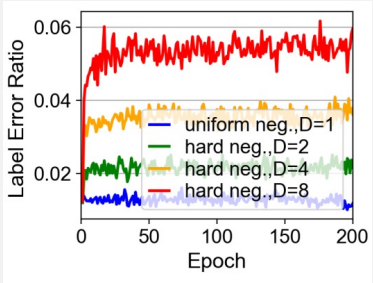

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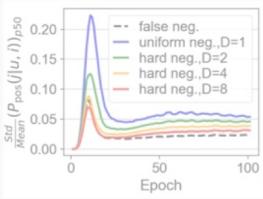
False Negatives Are More Stable



$Normalized\ Variance = \frac{Std(P_{pos})}{Mean(P_{pos})}$

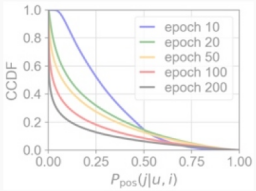
- As D increases, the Label Error Ratio consistently increases
  - Aggressive hard-negative mining increases false-negative contamination

Understanding False Negatives

SRNS: Hard but Reliable Negatives

- **Prediction variance provides an additional reliability signal**

Only a Few Negatives Are Hard



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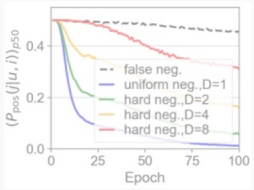

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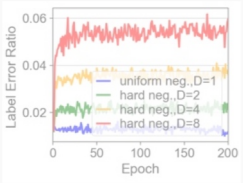

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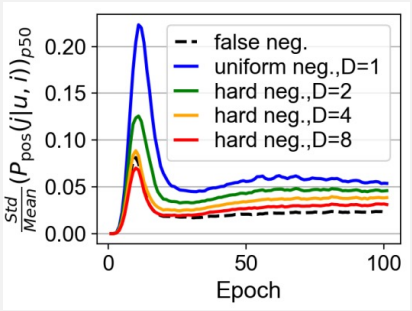
False Negatives Are More Stable



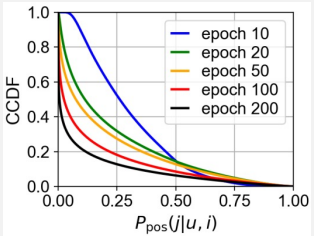
$$Normalized\ Variance = \frac{Std(P_{pos})}{Mean(P_{pos})}$$

- False negatives show lower prediction variance than true negative samples
  - Their predictions remain relatively stable throughout training

Understanding False Negatives (Cont.)

SRNS: Hard but Reliable Negatives

Only a Few Negatives Are Hard



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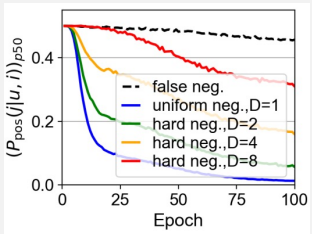
False Negatives Also Look Hard



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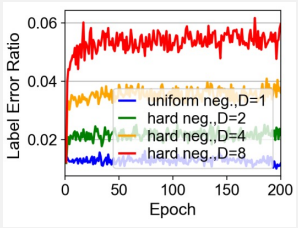

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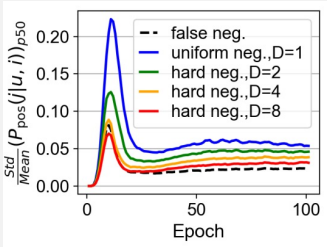
False Negatives Are More Stable



$$\textit{Normalized Variance} = \frac{Std(P_{pos})}{Mean(P_{pos})}$$

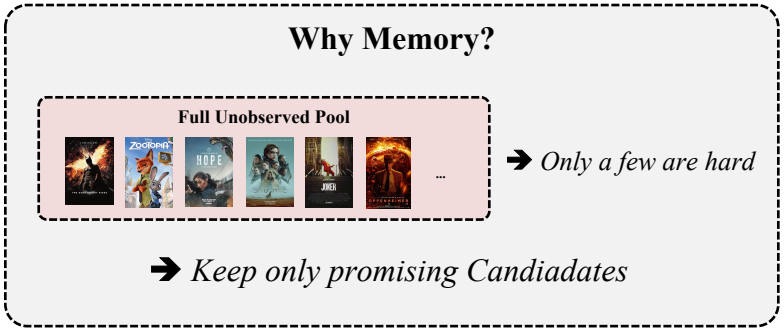
Current Score ➔ **Hardness**

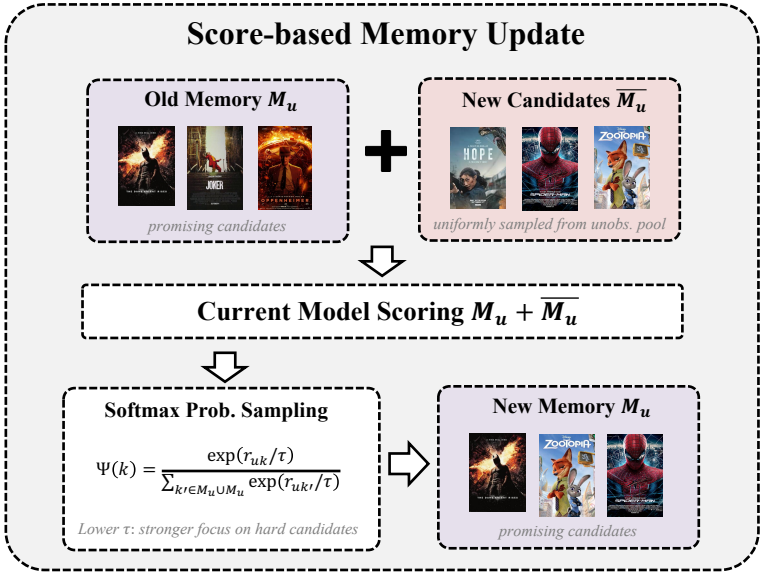
Prediction History ➔ **Reliability**

Remember the Hard Ones

SRNS: Hard but Reliable Negatives

- **Maintain a small memory of promising negatives**



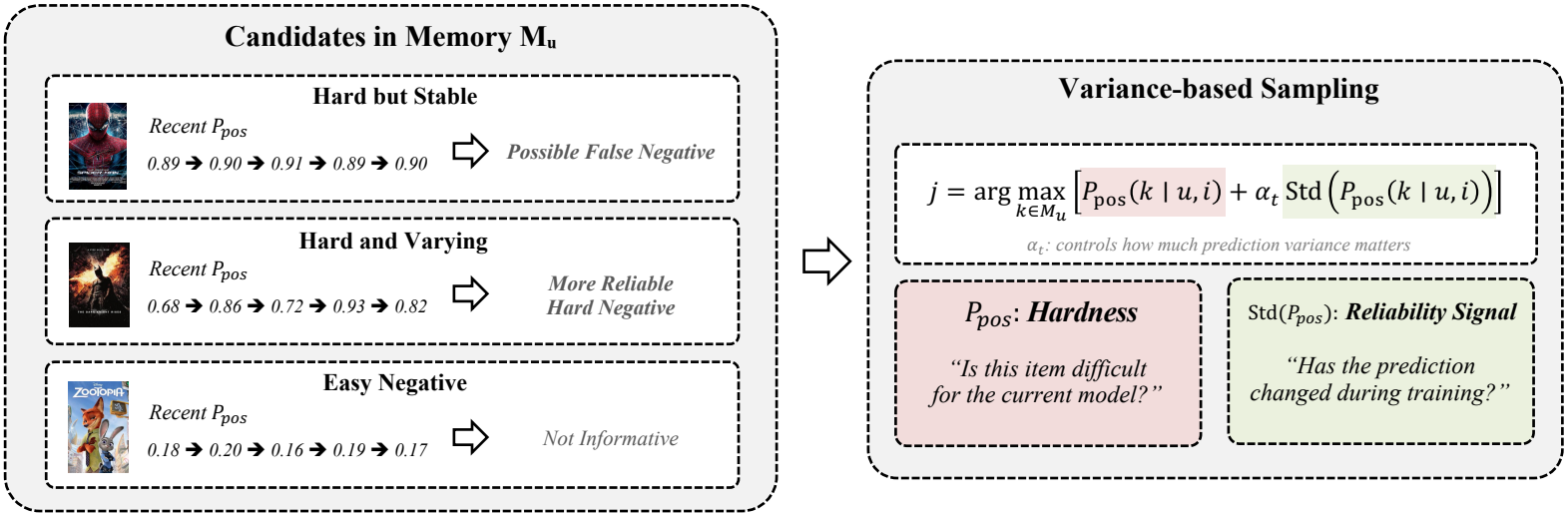


- Remember promising negatives instead of rediscovering them from the full pool
- **But high-score memory can still contain false negatives. How should we choose one reliably?**

Hard, But Reliable

SRNS: Hard but Reliable Negatives

- **Select negatives using both hardness and prediction stability**

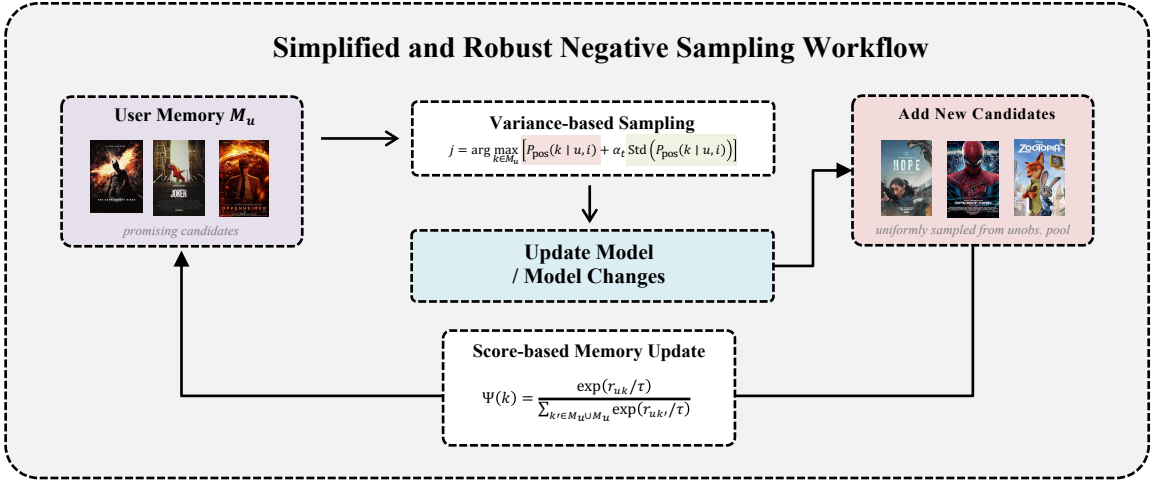


- Do not simply choose the hardest negative — prefer one that is hard and less likely to be false.

SRNS Workflow

SRNS: Hard but Reliable Negatives

- Alternating between robust sampling and dynamic memory update



Warm Start

$$\alpha_t = \alpha \min\left(\frac{t}{T_0}, 1\right)$$

Early: *Hardness first*
Later: *Hardness + Reliability*

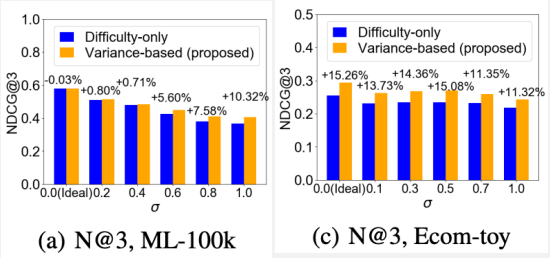
Does Reliability Help?

SRNS: Hard but Reliable Negatives

- **Variance-aware sampling improves robustness and recommendation quality**

Robustness to False Negatives

Category	Dataset	User	Item	Train	Validation	Test	$\mathcal{F}$ (Noise)
Synthetic noisy dataset	Movielens-100k	942	1,447	44,140	-	11,235	5,509
	Ecommerce-toy	1,000	2,000	60,482	-	14,612	3,246



(a) N@3, ML-100k (c) N@3, Ecom-toy

Variance-based sampling remains more robust as false-negative noise increases

Real-world Performance

Method	Movielens-1m			Pinterest			Ecommerce		
	N@1	N@3	R@3	N@1	N@3	R@3	N@1	N@3	R@3
ENMF	0.1846	0.3021	0.3882	0.2594	0.4144	0.5284	0.1317	0.2095	0.2670
Uniform	0.1744	0.2846	0.3663	0.2586	0.4136	0.5276	0.1265	0.2057	0.2640
NNCF	0.0829	0.1478	0.1971	0.2292	0.3699	0.4735	0.0833	0.1420	0.1855
AOBPR	0.1802	0.2905	0.3728	0.2596	0.4165	0.5319	0.1293	0.2108	0.2710
IRGAN	0.1755	0.2877	0.3708	0.2587	0.4143	0.5282	0.1275	0.2065	0.2648
RNS-AS	0.1823	0.2932	0.3754	0.2690	0.4233	0.5359	0.1335	0.2131	0.2714
AdvIR	0.1790	0.2941	0.3792	0.2689	0.4235	0.5363	0.1357	0.2141	0.2719
SRNS	**0.1933**	**0.3070**	**0.3912**	**0.2891**	**0.4391**	**0.5486**	**0.1471**	**0.2256**	**0.2833**
	4.71%	1.62%	0.77%	7.47%	3.68%	2.29%	8.40%	5.37%	4.19%

- Current score finds informative negatives, Prediction history reduces the risk of false negatives

On the Theories Behind Hard Negative Sampling for Recommendation

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Beyond Faster Convergence

DNS+: Why Does Hard Negative Sampling Work?

❖ **Is faster convergence enough to explain HNS?**

Conventional view

Hard Negative ➔ Larger Gradient
➔ Faster Convergence

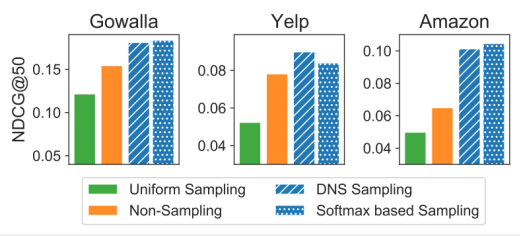
This explains training efficiency —
but does it explain better final accuracy?

BPR Loss with Sampling Distribution

$$\min_{\theta} \sum_{c \in C} \sum_{i \in \mathcal{I}_c^+} E_{j \sim P_{ns}(j|c)} \left[\ell \left(r(c,i|\theta) - r(c,j|\theta) \right) \right]$$

Negative Sampling Distribution changes
the effective training objective

An Unexpected Observation



HNS > Non-Sampling > Uniform

*But Non-sampling uses **ALL** negatives*

❑ HNS may change not only how fast we optimize, but what we optimize

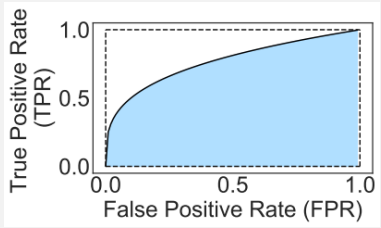
➔ **What does Hard Negative Sampling actually optimize?**

Focus on the Top

DNS+: Why Does Hard Negative Sampling Work?

- **Top-K recommendation does not care about every ranking error equally**

AUC: Evaluate the Entire Ranking

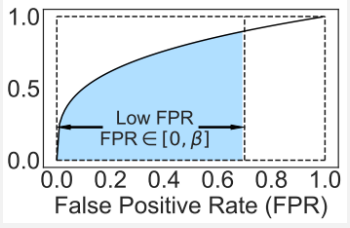


$$AUC = P\left(s(u, i^+) > s(u, j^-)\right)$$

A randomly chosen positive should outrank a randomly chosen negative

Recommendation is evaluated at the top of the list

OPAUC: Focus on the Top



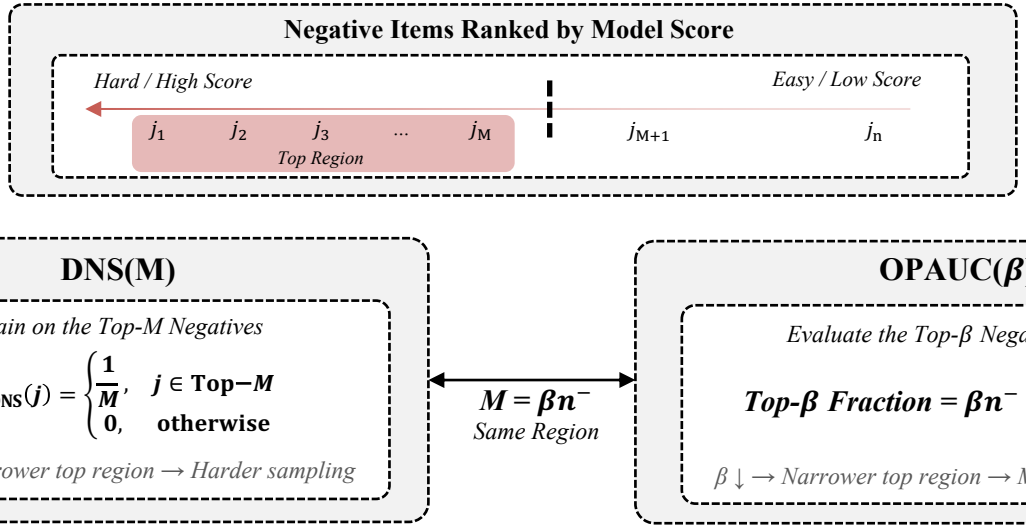
$$OPAUC(\beta) = \text{AUC over FPR} \in [0, \beta]$$
$$(OPAUC(1) = AUC)$$

Low FPR corresponds to a high score threshold
→ Only the highest-scored negatives are considered

- AUC considers the whole negative population, while OPAUC focuses on top-ranked high-scored negatives

Why Does HNS Optimize OPAUC?

❖ Both DNS and OPAUC focus on the same top-ranked negative region



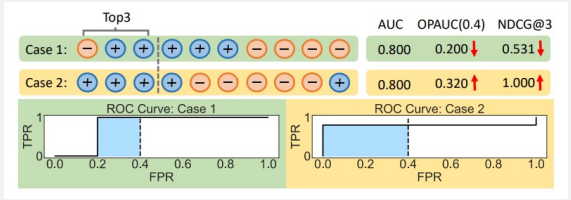
- ❑ BPR + DNS ≡ OPAUC(β) surrogate (Formal proof via CVaR-DRO)
- ❑ DNS shifts learning toward the ranking region that matters most at the top

Why OPAUC Matches Top-K

DNS+: Why Does Hard Negative Sampling Work?

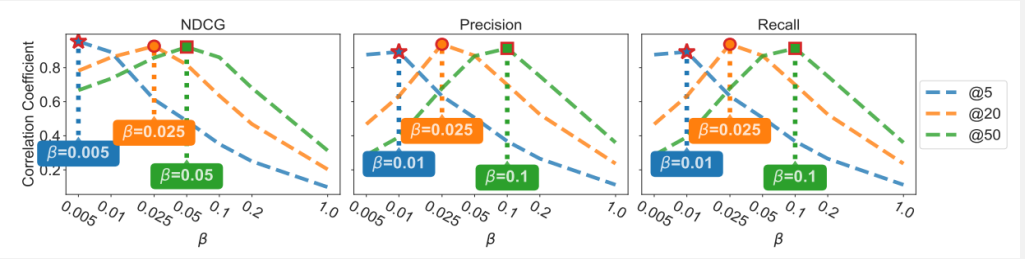
❖ **The same overall ranking quality can produce very different Top-K results**

Same AUC, Different Top-K



AUC may miss where ranking errors occur, while OPAUC is sensitive to errors near the top

Correlation with Top-K Metrics



OPAUC achieves stronger correlation with NDCG, Precision, and Recall than AUC

$K \downarrow \rightarrow \beta \downarrow$

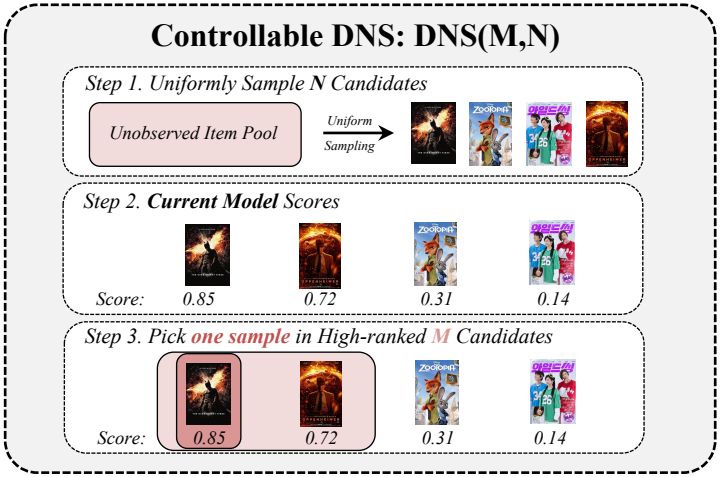
Smaller K favors more top-focused objective

- AUC measures global pairwise ranking quality, regardless of where the errors occur
- OPAUC emphasizes errors among high-ranked negatives
  - which are exactly the errors that affect Top-K recommendation
- **As the target K becomes smaller, the optimal OPAUC region also becomes narrower**

How Hard Should It Be?

DNS+: Why Does Hard Negative Sampling Work?

❖ Optimal hardness depends on the target Top-K metric



Top-K Determines the Right Hardness

$K \downarrow \rightarrow \beta \downarrow$

$\rightarrow$ *Harder Negatives*

- Hardness should be tuned to the ranking region emphasized by the target metric
- Smaller K focuses evaluation closer to the top, requiring harder negative samples

Does Controlled Hardness Help?

DNS+: Why Does Hard Negative Sampling Work?

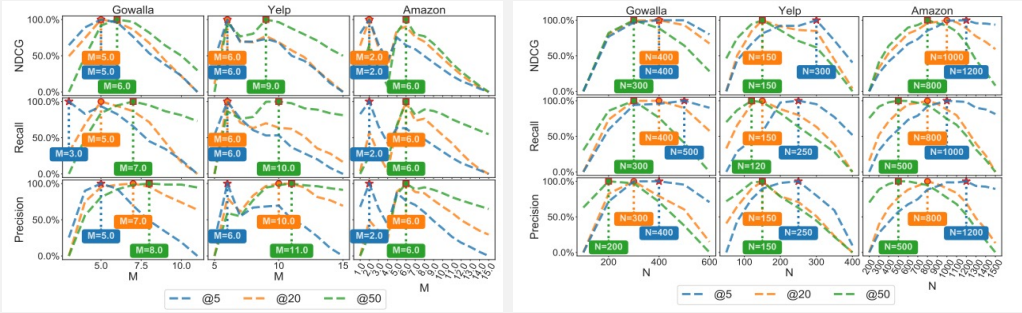
- **Controllable hardness improves performance and follows the Top-K guideline**

Controllable HNS Improves Performance

Method	Gowalla		Yelp		Amazon	
	NDCG@50	Recall@50	NDCG@50	Recall@50	NDCG@50	Recall@50
BPR	0.1216	0.2048	0.0524	0.1083	0.0499	0.1171
AOBPR	0.1385	0.2417	0.0677	0.1346	0.0563	0.1303
WARP	0.1248	0.2240	0.0636	0.1332	0.0542	0.1267
IRGAN	0.1443	0.2242	0.0695	0.1367	0.0627	0.1395
Kernel	0.1399	0.2264	0.0658	0.1315	0.0700	0.1495
DNS	0.1412	0.1839	0.0693	0.1425	0.0615	0.1378
PRIS(U)	0.1334	0.2217	0.0639	0.1273	0.0607	0.1377
PRIS(P)	0.1385	0.2282	0.0673	0.1342	0.0697	0.1463
AdaSIR(U)	0.1489	0.2500	0.0732	0.1523	0.0731	0.1505
AdaSIR(P)	0.1519	0.2516	0.0731	0.1525	0.0740	0.1534
DNS(M, N)	0.1811**	0.2989**	**0.0899****	**0.1774****	0.1014**	0.1833**
Softmax-v(ρ, N)	**0.1837****	**0.2993****	0.0840**	0.1690**	**0.1046****	**0.1937****

Controlling hardness consistently outperforms fixed hard-negative sampling

Smaller K Prefers Harder Negatives

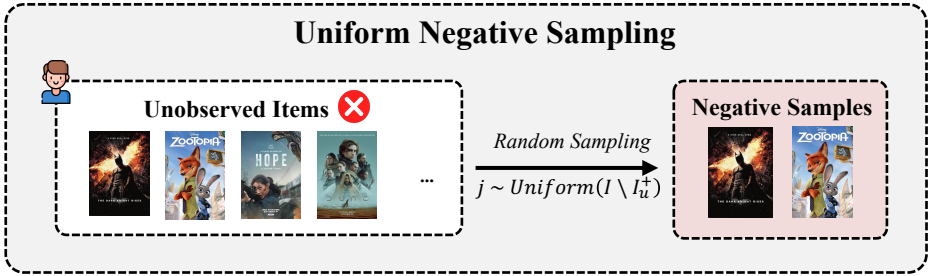


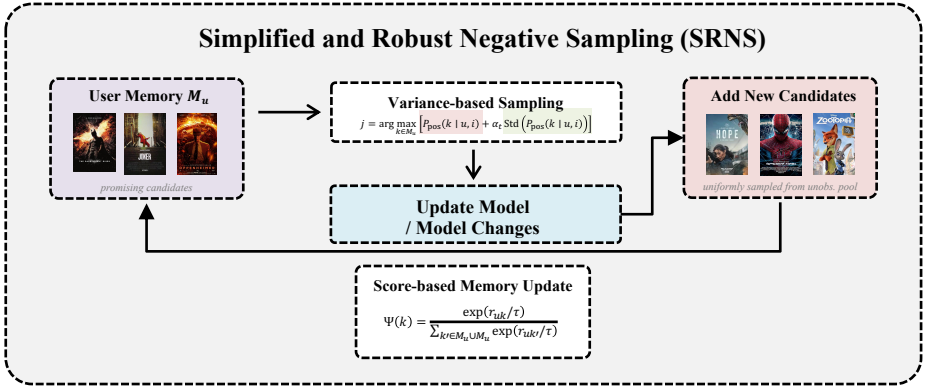
$K \downarrow \rightarrow M^* \downarrow, N^* \uparrow$

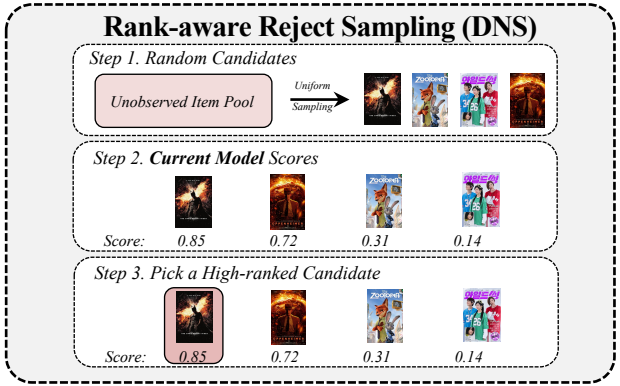
➔ *Harder Sampling*

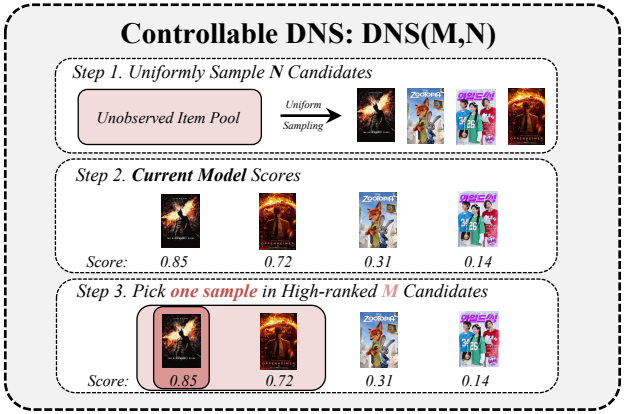
- Hardness should be controlled according to the ranking region emphasized by the target metric

Conclusion


Uniform Negative Sampling
Unobserved Items
...
Random Sampling
$j \sim Uniform(I \setminus I_u^+)$
Negative Samples


Simplified and Robust Negative Sampling (SRNS)
User Memory M_u
promising candidates
Variance-based Sampling
$j = \arg\max_{k \in M_u} \left[P_{pos}(k \mid u, i) + \alpha_s \operatorname{Std}\left(P_{pos}(k \mid u, i)\right) \right]$
Update Model
/ Model Changes
Add New Candidates
uniformly sampled from unobs. pool
Score-based Memory Update
$\Psi(k) = \frac{\exp(r_{uk}/\tau)}{\sum_{k' \in M_u \cup M_u} \exp(r_{uk'}/\tau)}$


Rank-aware Reject Sampling (DNS)
Step 1. Random Candidates
Unobserved Item Pool
Uniform Sampling
Step 2. Current Model Scores
Score: 0.85 0.72 0.31 0.14
Step 3. Pick a High-ranked Candidate
Score: 0.85 0.72 0.31 0.14


Controllable DNS: DNS(M,N)
Step 1. Uniformly Sample N Candidates
Unobserved Item Pool
Uniform Sampling
Step 2. Current Model Scores
Score: 0.85 0.72 0.31 0.14
Step 3. Pick one sample in High-ranked M Candidates
Score: 0.85 0.72 0.31 0.14


THANKS

Appendix (SRNS Time Complexity Table)

rank sorted by score, pop_j is the j's item popularity, B is the mini-batch size, T is the time complexity of computing an instance score, E is the epoch of lazy-update, and $\mathcal{F}$ denotes false negative.

	$p_{\text{ns}}(j\|u)$	Optimization	Time Complexity	Robustness
Uniform [33]	$\text{Uniform}(\{j \notin \mathcal{R}_u\})$	SGD (from scratch)	$O(BT)$	×
NNCF [10]	$\propto (\text{pop}_j)^{0.75}$	SGD (from scratch)	$O(B^2T)$	×
AOBPR [32]	$\propto \exp(-\text{rk}(j\|u)/\lambda)$	SGD (from scratch)	$O(BT)$	×
IRGAN [37]	learned $\bar{p}_{\text{ns}}(j\|u)$ (GAN)	REINFORCE (pretrain)	$O(B\|\mathcal{I}\|T)$	×
AdvIR [28]	learned $\bar{p}_{\text{ns}}(j\|u)$ (GAN)	REINFORCE (pretrain)	$O(BS_1T)$	×
SRNS (proposed)	variance-based (see (4))	SGD (from scratch)	$O(\frac{B}{E}(S_1+S_2)T)$	√

DRO as a Bridge to Hard Negative Sampling

- **DRO asks for the worst-case sampling distribution**
  - Worst-case Reweighting
    - $\max_{Q} \mathbb{E}_{j\sim Q}\left[L(c,i,j)\right] \quad \text{s.t.} \quad D_{\text{CVaR}}(Q\|P_0) \le \rho$
      - Choose a negative-sampling distribution Q that maximizes the current loss, but cannot move too far from P_0 (Uniform Distribution)
  - CVaR constraint limits each probability
    - $D_{\text{CVaR}}(Q\|P_0) = \sup_j \log\frac{Q(j)}{P_0(j)} \le \rho,\ Q(j) \le e^{\rho}P_0(j) = \frac{e^{\rho}}{n_-}$
  - Then what is the worst-case Q∗?
    - $Q^*(j) = \begin{cases} \frac{1}{M}, & j \in \text{TopM}(L) \\ 0, & \text{otherwise} \end{cases},\ M = \frac{1}{e^{\rho}/n_-} = n_- e^{-\rho}$

DRO as a Bridge to Hard Negative Sampling (Cont.)

- **Why is this DNS?**
  - BPR Loss: $L(c,i,j) = \log\left(1 + \exp\left(r_{cj} - r_{ci}\right)\right)$, $r_{cj} \uparrow \Leftrightarrow L(c,i,j) \uparrow$
  - Negative score ranking = BPR loss ranking ➔ **TopM(ScoreRank) = TopM(Loss)**

- **DRO <-> HNS (DNS)**
  - $Q^*(j) = \begin{cases} \frac{1}{M}, & j \in \text{TopM}(L) \\ 0, & \text{otherwise} \end{cases}$, **TopM(L) = TopM(r)**
  - ➔ $Q^*(j) = \begin{cases} \frac{1}{M}, & j \in \text{TopM}(\text{r}) \\ 0, & \text{otherwise} \end{cases} = \boldsymbol{P_{\text{DNS}}(j)}$

From OPAUC Metric to Trainable Objective

How a partial ranking metric becomes a BPR-style loss

- $OPAUC(\beta) = \frac{1}{|\mathcal{C}|}\sum_{c\in\mathcal{C}} \int_0^{\beta} TPR_{c,\theta}\left[FPR_{c,\theta}^{-1}(s)\right]ds$
  - OPAUC only evaluates the low-FPR region

- Convert it to a finite-data pairwise estimator
  - $\widehat{OPAUC}(\beta) = \frac{1}{|\mathcal{C}|}\sum_{c\in\mathcal{C}} \frac{1}{n_+ n_-}\sum_{i\in\mathcal{I}_c^+}\sum_{j\in S_{j_c^-}^{\downarrow}[1,n_-\beta]} \mathbb{I}\left(r_{ci} > r_{cj}\right)$
  - negative items sorted by score, taking only the top $n_-\beta$

- Replace the indicator with a surrogate loss
  - $L(c,i,j) = \ell\left(r_{ci} - r_{cj}\right)$
  - $\min_{\theta} \frac{1}{|\mathcal{C}|}\sum_{c\in\mathcal{C}} \frac{1}{n_+}\sum_{i\in\mathcal{I}_c^+} \frac{1}{n_-\beta}\sum_{j\in S_{j_c^-}^{\downarrow}[1,n_-\beta]} L(c,i,j)$

Why Does DRO Optimize OPAUC?

- **OPAUC and CVaR-DRO optimize the same top-ranked negative region**
  - OPAUC as a trainable objective
    - $\mathcal{L}_{\mathrm{OPAUC}}(\beta) = \frac{1}{n\_\beta}\sum_{j\in \mathrm{Top}_{n\_\beta}} L\,(c,i,j)$
    - Average BPR-style loss over only the Top-β fraction of negatives
  - Recall the CVaR-DRO
    - $\mathcal{L}_{DRO} = \frac{1}{M}\sum_{j\in Top_M} L\,(c,i,j)$
  - LEMMA: $\beta = e^{-\rho}$
    - $M = n\_e^{-\rho} = n\_\beta$

Why Does DRO Optimize OPAUC? (Cont.)

OPAUC Objective

$$\frac{1}{\beta n_-} \sum_{j \in Top_{\beta n_-}} L_j$$

CVaR-DRO Objective

$$\frac{1}{M} \sum_{j \in Top_M} L_j$$

$$M = n_- e^{-\rho} = \beta n_-$$

$$\frac{1}{\beta n_-} \sum_{j \in Top_{\beta n_-}} L_j = \frac{1}{M} \sum_{j \in Top_M} L_j \Rightarrow \text{CVaR-DRO} \equiv \text{OPAUC}(\beta) \text{ surrogate}$$

Why Does OPAUC Connect to Top-K?

Top-K Quality

$$Recall@K = \frac{m}{N_+}, \quad Precision@K = \frac{m}{K}$$

$$m = \#\,positives\ in\ Top\text{-}K$$

Fix m, Find Extremes

Best arrangement

$$\underbrace{+\cdots+}_{m}\underbrace{-\cdots-}_{K-m} \mid \underbrace{+\cdots+}_{N_+-m}\underbrace{-\cdots-}_{N_--K+m}$$

$$\boldsymbol{OPAUC_{max}(m)}$$

Worst arrangement

$$\underbrace{-\cdots-}_{K-m}\underbrace{+\cdots+}_{m} \mid \underbrace{-\cdots-\cdots}_{m}$$

$$\boldsymbol{OPAUC_{\min}(m)}$$

$$\boldsymbol{OPAUC_{\min}(m)} \le A \le \boldsymbol{OPAUC_{max}(m)}$$

A: Actual ranking's OPAUC

Solve for m under fixed A

$$\boldsymbol{m_{min}(A) \le m \le m_{max}(A)}$$

$$\frac{m_{\min}}{N_+} \le Recall@K \le \frac{m_{\max}}{N_+}$$

$$\frac{m_{\min}}{K} \le Precision@K \le \frac{m_{\max}}{K}$$